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1 KERALA STARTUP BUILDING HUMANOID ROBOTS FROM SCRATCH

An Indian startup is building humanoid robots in-house. The robots are trained like workers, tested in controlled spaces, corrected using feedback, and improved through repeated practice.

Three friends from Kerala—Athil Krishna, Akhil K. Haridasan, and Sarath S.—began working on humanoid robots for defence and real-world applications. After months of developing proposals, they submitted multiple ideas to the government in 2019, highlighting the importance of humanoid robots for the army. In 2020, the BSF and PMO responded, with the BSF expressing interest in a demo. However, the Covid pandemic disrupted progress. Without a working prototype and having already returned to Kerala, the team could not present the demonstration.

They resumed work in 2022 and founded iHub Robotics (Innovation Hub for Robotics) to develop robotics solutions.

The company builds humanoid robots in India. Its first product, Tara Generation 1, launched in 2024 as a service humanoid robot capable of speaking more than 100 languages, displaying human-like behaviour, and autonomously navigating spaces up to 400,000 square feet (approx. 37,160 square metres). Tara is available in variants such as Tara Greet, Tara Lean, and Tara Care for applications across healthcare, education, offices, and general services.

After Tara, the company built



Athil Krishna, Founder & CEO, iHub Robotics

its second major system, Daksha, a general-purpose industrial humanoid robot. Daksha is designed for heavy industrial environments and can lift payloads of up to 25kg, making it one of the strongest humanoid systems in its category. It can also move vertically, operating at heights of up to 8-10 feet (2.4-3 metres) for industrial tasks.

The company is also developing India's first indigenous 'Viveka Decision Core,' an advanced brain-like orchestration system that powers these robots and controls decision-making and movement.

"We are working on an advanced AI called VLA (Vision Language

Action). Unlike LLMs (Large Language Models), which mainly handle text and content generation, VLA goes one step further into 'prompt to action.' It connects vision (eyes), language (understanding), and action (movement). In simple terms, it helps robots see, understand, and physically act in the real world. A related concept is VLN (Vision Language Navigation), where robots can understand environments and choose paths on their own, like navigating multiple routes in an unknown space," says Athil.

Speaking about the design challenges, Athil adds,

"Designing a humanoid robot is complex because it functions like a human body with multiple mechanical and kinematic systems that require deep research and repeated prototyping before finalisation. The hardware is also challenging, with many joints and motors, each needing precise control, feedback, and voltage management. Integration with high-end computing boards adds another layer of difficulty. Over the last three years, we have developed our own hardware circuits and control boards, enabling faster development and greater independence. Most components today, including body systems, battery management,